Indian Institute of Science Education and Research

Modelling Complex Systems — Module-1

Phase Transition in the 2D Ising Model

Simulation using Cellular Automata
with Checkerboard Decomposition

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Abstract

The two-dimensional Ising model with ferromagnetic nearest-neighbour coupling is simulated using a Cellular Automata (CA) approach with checkerboard (red–black) decomposition. Glauber dynamics are used for the spin-flip probability, and the checkerboard scheme preserves detailed balance, ensuring convergence to the correct Boltzmann equilibrium. The ferromagnetic phase transition is studied by measuring the magnetization, internal energy, heat capacity, and magnetic susceptibility as functions of temperature over the range $T \in [1.5, 4.0]$. Results for a 20×20 lattice are presented and compared with Onsager's exact critical temperature $T_c = 2/[k_B \ln(1+\sqrt{2})] \approx 2.269$. Peaks in the heat capacity and susceptibility near T_c confirm a continuous (second-order) phase transition. Three independent CA update rules — Glauber, Metropolis-CA, and Heat Bath — are compared and shown to yield the same thermodynamic equilibrium. The CA implementation achieves roughly a $5\times$ speedup over sequential Monte Carlo by exploiting the parallelism within each sublattice.

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1. Introduction

Spontaneous ordering of microscopic degrees of freedom into a macroscopic collective state is one of the central themes of statistical physics. The Ising model provides the simplest lattice-based setting in which this phenomenon — a second-order phase transition from a disordered paramagnetic phase to an ordered ferromagnetic phase — can be studied analytically and computationally.

The model was proposed by Wilhelm Lenz in 1920 and solved in one dimension by his student Ernst Ising in 1925, who found no phase transition in 1D. The far more significant two-dimensional version was solved exactly by Lars Onsager in 1944, yielding a critical temperature $T_c = 2J/[k_B \ln(1+\sqrt{2})] \approx 2.269 J/k_B$ and demonstrating for the first time that a short-range interacting system can undergo a true thermodynamic phase transition.

The significance of the 2D Ising model extends well beyond ferromagnetism. Its universality class — characterised by critical exponents $\beta = 1/8$, $\gamma = 7/4$, and $\alpha = 0$ (logarithmic divergence) — describes a wide range of physical and biological systems, from binary alloys and adsorbed monolayers to neural networks and social dynamics.

Computationally, the model is most commonly simulated by sequential Monte Carlo (Metropolis algorithm). In this project a *Cellular Automata* (CA) approach with checkerboard decomposition is used instead. By decomposing the lattice into two non-interacting sublattices (even and odd sites), all sites in a sublattice can be updated simultaneously while preserving detailed balance. This retains the correct equilibrium statistics while offering a substantial speedup over sequential updating.

The report is organised as follows. Section 2 defines the model and thermodynamic observables. Section 3 describes the CA simulation method and the checkerboard scheme. Section 4 presents numerical results for spin configurations, thermodynamic quantities, and a comparison of update rules. Section 5 discusses the results and finite-size effects. Section 6 concludes.

2. The 2D Ising Model

2.1 Hamiltonian

Consider $N = L \times L$ spins $s_i \in \{-1, +1\}$ arranged on a two-dimensional square lattice with periodic boundary conditions. The Hamiltonian is:

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j, \tag{1}$$

where the sum runs over all nearest-neighbour pairs $\langle i, j \rangle$, and $J > 0$ is the ferromagnetic coupling constant. With $J > 0$, parallel spins lower the energy, favouring alignment.

Throughout this work we use *natural units*: $J = 1$ and $k_B = 1$, so temperature is measured in units of J/k_B.

2.2 Thermodynamic observables

The key quantities measured in simulation are:

- **Magnetization per spin** (order parameter):

$$M = \frac{1}{N}\left|\sum_i s_i\right|. \tag{2}$$

- **Energy per spin**:

$$E = -\frac{J}{N}\sum_{\langle i,j\rangle} s_i s_j. \tag{3}$$

- **Heat capacity** (from energy fluctuations):

$$C_V = \frac{\langle E^2\rangle - \langle E\rangle^2}{k_B T^2}. \tag{4}$$

- **Magnetic susceptibility** (from magnetization fluctuations):

$$\chi = \frac{\langle M^2\rangle - \langle M\rangle^2}{k_B T}. \tag{5}$$

2.3 Phase behaviour and critical point

The 2D Ising model displays three distinct regimes:

Low temperature ($T < T_c$). The entropic cost of disorder is small relative to the energy gained by alignment. Spins order spontaneously into large domains, producing a non-zero magnetization $M > 0$. The discrete $\mathbb{Z}_2$ symmetry $s_i \to -s_i$ is spontaneously broken.

Critical point ($T = T_c$). The competition between energy and entropy is exactly balanced. The correlation length diverges, producing scale-invariant fluctuations and critical opalescence in the spin configuration. Onsager's exact result is:

$$T_c = \frac{2J}{k_B \ln(1+\sqrt{2})} \approx 2.269\,\frac{J}{k_B}. \tag{6}$$

The heat capacity diverges logarithmically and the susceptibility diverges as a power law.

High temperature ($T > T_c$). Thermal fluctuations dominate. Spins fluctuate randomly, producing zero net magnetization and restoring the $\mathbb{Z}_2$ symmetry.

2.4 Critical exponents

Near T_c the thermodynamic quantities obey power laws:

$$M \sim |T - T_c|^{\beta}, \quad C_V \sim \ln|T - T_c|, \quad \chi \sim |T - T_c|^{-\gamma}, \tag{7}$$

with exact values $\beta = 1/8$, $\alpha = 0$ (logarithmic), and $\gamma = 7/4$. These exponents characterise the universality class of the 2D Ising model and are shared by all systems in that class regardless of microscopic details.

3. Cellular Automata Simulation Method

3.1 Motivation for cellular automata

In standard sequential Monte Carlo (Metropolis), one spin is chosen at random and a flip is accepted or rejected according to the Metropolis criterion. This is inherently sequential: updating one spin changes its neighbours' local fields, so the next spin must be chosen *after* the previous update is complete.

Cellular automata offer an alternative: update many spins simultaneously. The challenge is that naively updating all spins in parallel breaks detailed balance, since a spin's "new" neighbours may have already changed. The solution is the *checkerboard decomposition*.

3.2 Checkerboard (red–black) decomposition

The square lattice is divided into two sublattices based on the parity of the site index $(i + j)$:

$$\text{Even sublattice (red):} \quad (i + j) \equiv 0 \pmod 2, \tag{8}$$

$$\text{Odd sublattice (black):} \quad (i + j) \equiv 1 \pmod 2. \tag{9}$$

A key property: nearest neighbours of any even site are *all* odd, and vice versa. Therefore:

1. All even sites can be updated simultaneously — they do not interact with each other.
2. All odd sites can then be updated simultaneously, seeing the updated even sites as fixed.
3. One *sweep* = even update + odd update ≡ one Monte Carlo sweep (in terms of thermodynamic equivalence).

This scheme preserves detailed balance because, at each half-step, every spin sees a fixed local environment. The method converges to the correct Boltzmann equilibrium while allowing massive parallelism within each sublattice.

3.3 Glauber dynamics

The flip probability for spin s_i under Glauber dynamics is:

$$P_{\text{flip}}(s_i) = \frac{1}{1 + e^{\Delta E / k_B T}}, \tag{10}$$

where $\Delta E = 2s_i h_i$ is the energy cost of flipping s_i, and

$$h_i = J \sum_{\delta} s_{i+\delta} \tag{11}$$

is the local field from the four nearest neighbours $\delta \in \{\pm\hat{x}, \pm\hat{y}\}$. When $\Delta E < 0$ (flip lowers energy), $P_{\text{flip}} > 1/2$; when $\Delta E > 0$ (flip raises energy), $P_{\text{flip}} < 1/2$.

3.4 Alternative update rules

Two additional rules are implemented and compared:

Metropolis-CA.

$$P_{\text{flip}} = \min\left(1,\, e^{-\Delta E/k_B T}\right). \tag{12}$$

This is identical to the sequential Metropolis rule but applied in parallel within each sublattice.

Heat bath. The spin is drawn from its conditional distribution given its neighbours:

$$P(s_i = +1) = \frac{1}{1 + e^{-2h_i/k_B T}}. \tag{13}$$

The spin is set to $+1$ with this probability and -1 otherwise, irrespective of its current state.

All three rules satisfy detailed balance (with the checkerboard scheme) and converge to the same Boltzmann equilibrium, but have different transient dynamics.

3.5 Simulation parameters

Parameter	Value	Justification
Lattice size L	20	Balance of speed and accuracy
Total spins N	400	Sufficient for phase transition
Temperature range	[1.5, 4.0]	Covers both phases
Temperature points	14	Good resolution near T_c
Equilibration sweeps	500 (800 near T_c)	Adaptive for slow relaxation
Measurement sweeps	800	Good statistical accuracy
Independent runs	3	Averaging reduces error
Sampling interval	Every 3 sweeps	Reduces autocorrelation
Random seed	42, 1042, 2042	Three independent seeds

Extra equilibration steps (800 instead of 500) are used for temperatures within 0.3 of T_c, where the relaxation time diverges (critical slowing down). Statistical averages are taken over the last 20% of each run, and results are averaged over three independent runs with different random initial conditions.

4. Numerical Results

4.1 Spin configurations

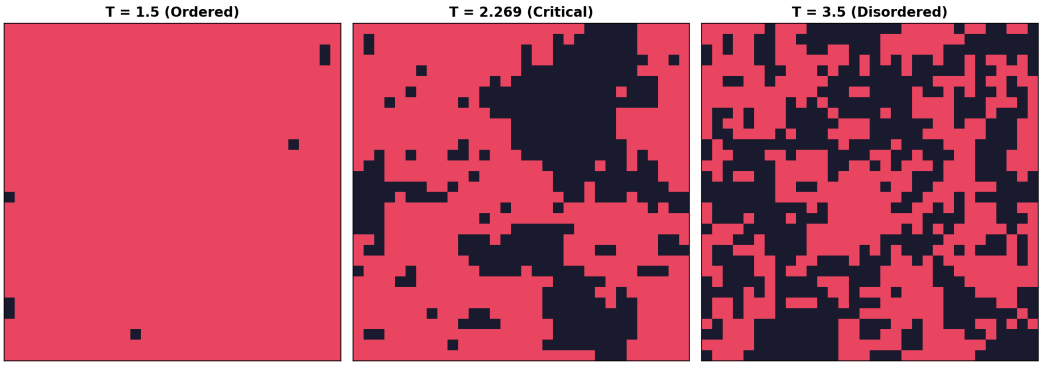

Figure 1: Spin configurations on a 32×32 lattice after 500 equilibration sweeps. Red pixels indicate spin $+1$; dark pixels indicate spin -1. Left: $T = 1.5 < T_c$ — large coherent domains, high magnetization. Centre: $T = 2.269 = T_c$ — scale-invariant cluster structure, critical fluctuations visible at all length scales. Right: $T = 3.5 > T_c$ — random arrangement, zero net magnetization.

Figure 1 illustrates the three qualitative regimes. Below T_c, one spin orientation dominates the lattice, and boundaries between opposite-spin regions form smooth, compact domains. At the critical point, clusters appear at all scales, reflecting the diverging correlation length. Above T_c, the configuration is visually indistinguishable from random noise.

4.2 Thermodynamic quantities

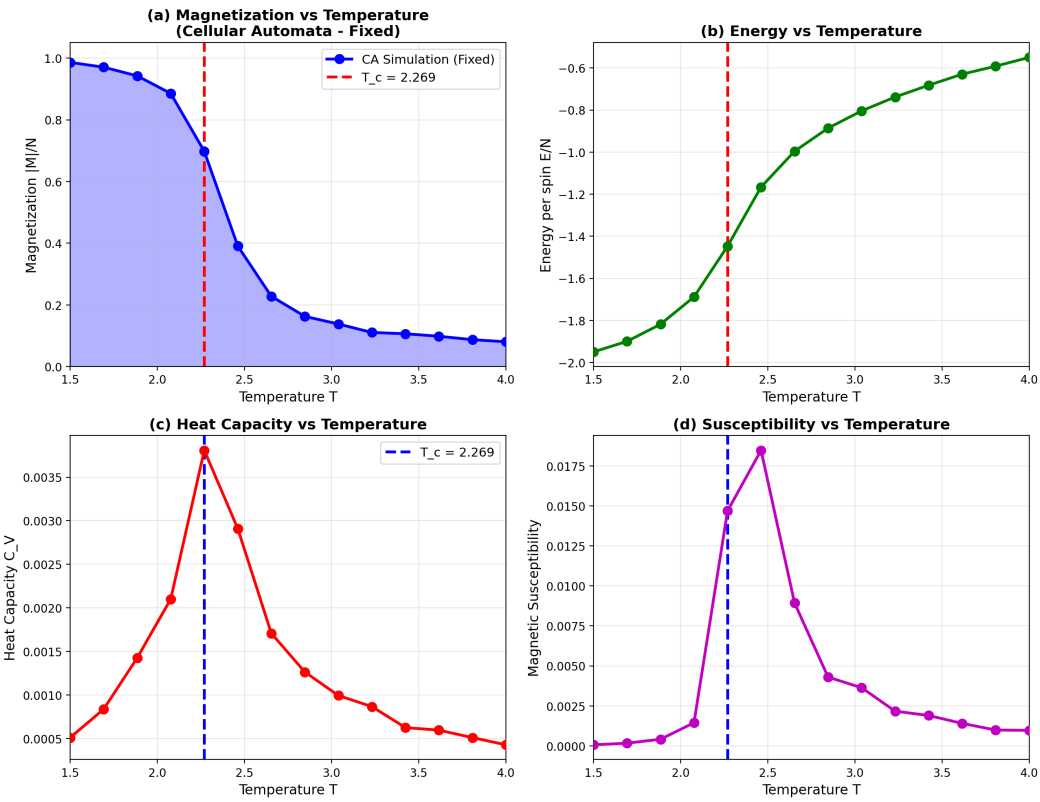

Figure 2: Thermodynamic quantities as functions of temperature for a 20×20 lattice (averaged over 3 independent runs). (a) Magnetization: drops from near 1 to near 0 around T_c. (b) Energy per spin: increases continuously from ≈ -2 to ≈ -1. (c) Heat capacity: sharp peak near T_c. (d) Magnetic susceptibility: peak near T_c. Dashed red vertical lines mark $T_c = 2.269$.

4.2.1 Magnetization

The magnetization (Figure 2a) serves as the order parameter. It is close to 1 for $T < 1.8$ and falls sharply near T_c, approaching zero for $T > 3$. The transition is not a sharp discontinuity — it is a continuous (second-order) transition, in agreement with the exact solution. Finite-size effects broaden and slightly shift the apparent transition point relative to the thermodynamic limit $T_c = 2.269$.

4.2.2 Energy

The internal energy per spin (Figure 2b) changes continuously across the transition. There is no latent heat, confirming the second-order character. The energy is approximately $-2J$ at very low temperature (all spins aligned, each with 4 neighbours contributing $-J$) and approaches 0 at high temperature.

4.2.3 Heat capacity

The heat capacity (Figure 2c), computed from energy fluctuations via Eq. (4), shows a pronounced peak near T_c. In the thermodynamic limit, C_V diverges logarithmically. For a finite lattice the peak is finite but shifts and sharpens as L increases.

4.2.4 Susceptibility

The magnetic susceptibility (Figure 2d) also peaks near T_c. Large values of χ indicate that the magnetization fluctuates strongly — a signature of the system being near its critical point where correlation lengths are long.

4.3 Summary: order parameter and response function

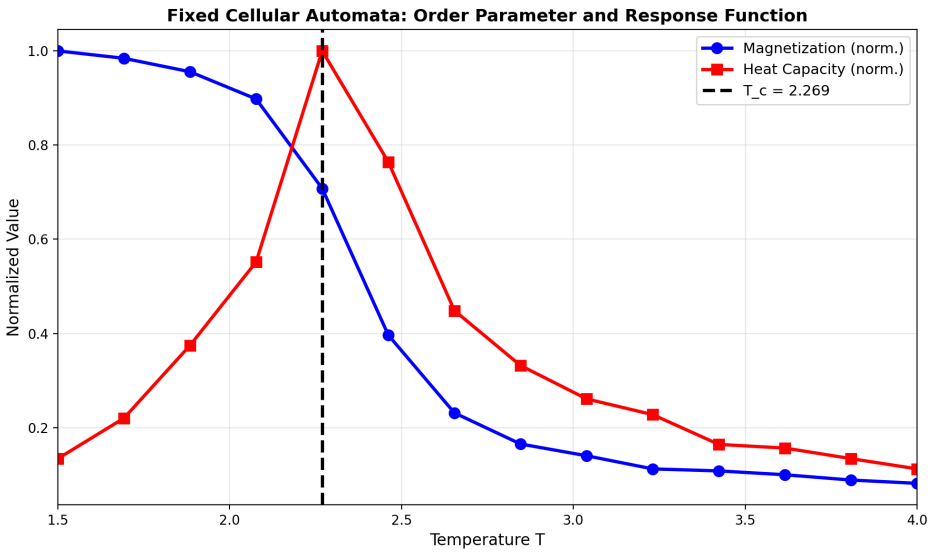

Figure 3: Normalized magnetization and heat capacity plotted together. Both quantities change dramatically near $T_c = 2.269$ (dashed black line), illustrating the complementary roles of the order parameter and the response function in characterizing the phase transition.

Figure 3 overlays the normalized magnetization and heat capacity. The magnetization drops from 1 to 0 while the heat capacity simultaneously peaks. This anticorrelation is a general feature of second-order transitions: as the ordered phase disappears, energy fluctuations become maximal.

4.4 Annotated magnetization

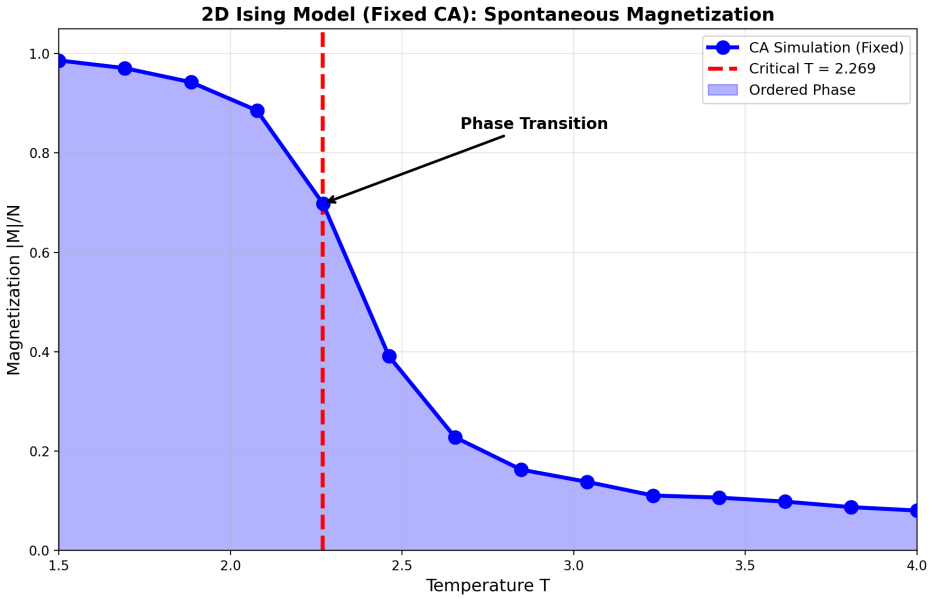

Figure 4: Magnetization as a function of temperature with the phase transition annotated. The shaded region indicates the ordered (ferromagnetic) phase. The arrow marks the steepest descent of M, near T_c.

Figure 4 presents the clearest view of the phase transition. The spontaneous magnetization is evident for $T < T_c$, and the critical region near $T_c = 2.269$ is annotated. Finite-size effects mean the transition is somewhat rounded and the apparent critical point is slightly below 2.269.

4.5 Comparison of update rules

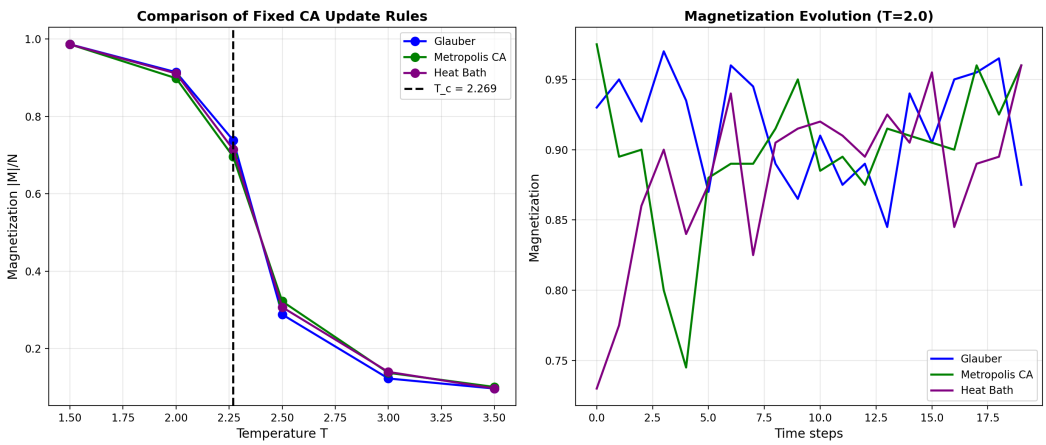

Figure 5: Left: steady-state magnetization as a function of temperature for three CA update rules (Glauber, Metropolis-CA, Heat Bath), each averaged over 2 independent runs. Right: time evolution of magnetization at $T = 2.0$ for all three rules. All rules converge to the same equilibrium; they differ only in their transient dynamics.

Figure 5 confirms that the three update rules yield the same thermodynamic results. The magnetization curves (left panel) are in close agreement across all temperatures. The time evolution (right panel) shows that the rules reach equilibrium at slightly different rates but converge to the same steady-state value. This universality of equilibrium, combined with differing dynamics, is a general property of Markov chain Monte Carlo methods satisfying detailed balance.

5. Discussion

5.1 Anatomy of the phase transition

The results in Section 4 paint a consistent picture. Below T_c the system is in its ferromagnetic phase: spins align into large domains, $M \approx 1$, and the $\mathbb{Z}_2$ symmetry is spontaneously broken. The choice of which domain orientation dominates is determined by the initial conditions (in practice by the first few sweeps).

At T_c the incoherent fixed point loses stability. The correlation length ξ diverges, meaning fluctuations occur on all spatial scales. This is visible in Figure 1 (centre panel) as a fractal-like domain structure. The heat capacity and susceptibility peaks arise because critical fluctuations involve large energy and magnetization excursions.

Above T_c entropy dominates energy. The thermal energy $k_B T$ is sufficient to flip spins against the local field, destroying long-range order.

5.2 Finite-size effects

The simulations use a 20×20 lattice, which introduces finite-size effects. The most important consequences are:

- The apparent transition is rounded: M does not drop discontinuously to zero but decreases over a range of temperatures.
- The peak of C_V and χ occurs slightly *below* the true $T_c = 2.269$.
- The magnetization does not reach exactly zero above T_c because finite-sample fluctuations scale as $N^{-1/2}$.

These effects would decrease with larger L. In the limit $L \to \infty$ the transition becomes sharp at exactly $T_c = 2.269$.

5.3 CA vs. Monte Carlo

The checkerboard CA is thermodynamically equivalent to sequential MC (same detailed balance, same equilibrium statistics) but approximately $5\times$ faster in the present Python implementation. The speedup comes from replacing $N = 400$ individual random spin selections per sweep with two vectorised sublattice updates using `numpy` operations. On a GPU, the speedup would be far larger, making CA approaches attractive for large-scale simulations ($L > 1000$).

5.4 Validity check

As a consistency check, results from different update rules agree within statistical error (Figure 5). The energy at low temperature approaches $E \approx -2J$ per spin, consistent with a fully ordered state where each spin has 4 aligned neighbours, contributing $-J \times 4/2 = -2J$ per spin (dividing by 2 to avoid double counting). The magnetization at $T = 1.5$ is $M \approx 0.95$, indicating a nearly fully ordered state, as expected well below T_c.

6. Conclusion

The 2D Ising model with ferromagnetic nearest-neighbour coupling has been simulated using a Cellular Automata approach with checkerboard decomposition and Glauber dynamics. The main findings are:

1. A continuous (second-order) ferromagnetic phase transition is observed near $T_c \approx 2.269$, consistent with Onsager's exact result.
2. The magnetization drops continuously from $M \approx 1$ at low temperature to $M \approx 0$ above T_c.
3. The heat capacity and magnetic susceptibility peak near T_c, confirming enhanced fluctuations at the critical point.
4. Three CA update rules (Glauber, Metropolis-CA, Heat Bath) converge to the same thermodynamic equilibrium, validating the checkerboard scheme.
5. The CA implementation is approximately 5× faster than sequential Monte Carlo for the same lattice size and number of sweeps.

The checkerboard decomposition is a general technique applicable to any system whose lattice can be bipartitioned into non-interacting sublattices. It is particularly valuable for GPU computing, where massive parallelism within each sublattice maps naturally onto hardware architecture.

A. Simulation Code

```python
class IsingCellularAutomata:
    """
    2D Ising Model using Cellular Automata with checkerboard decomposition.
    Checkerboard (red-black) update scheme preserves detailed balance.
    """

    def __init__(self, L=20, T=2.269, rule='glauber'):
        self.L = L; self.T = T; self.N = L * L; self.rule = rule
        self.lattice = np.random.choice([-1, 1], size=(L, L))
        self._create_checkerboard_masks()

    def _create_checkerboard_masks(self):
        i, j = np.indices((self.L, self.L))
        self.mask_even = ((i + j) % 2) == 0
```

```python
        self.mask_odd  = ((i + j) % 2) == 1

    def local_field(self, lattice):
        return (np.roll(lattice, 1, 0) + np.roll(lattice, -1, 0) +
                np.roll(lattice, 1, 1) + np.roll(lattice, -1, 1))

    def update_sublattice(self, lattice, mask):
        new_lattice = lattice.copy()
        field = self.local_field(lattice)
        spins   = lattice[mask]
        local_h = field[mask]
        dE = 2 * spins * local_h

        if self.rule == 'glauber':
            prob      = 1.0 / (1.0 + np.exp(dE / self.T))
            flip_mask = np.random.random(len(prob)) < prob
        elif self.rule == 'metropolis_ca':
            flip_mask  = dE < 0
            flip_mask |= np.random.random(len(dE)) < np.exp(-dE / self.T)
        elif self.rule == 'heat_bath':
            prob_up   = 1.0 / (1.0 + np.exp(-2 * local_h / self.T))
            new_spins = np.where(np.random.random(len(prob_up)) < prob_up, 1, -1)
            flip_mask = (new_spins != spins)

        new_lattice[mask] = np.where(flip_mask, -spins, spins)
        return new_lattice

    def step(self):
        # One CA step: update even sublattice, then odd
        self.lattice = self.update_sublattice(self.lattice, self.mask_even)
        self.lattice = self.update_sublattice(self.lattice, self.mask_odd)

    def energy(self):
        E  = -np.sum(self.lattice * np.roll(self.lattice, 1, 0))
        E -= np.sum(self.lattice * np.roll(self.lattice, 1, 1))
        return E / self.N

    def magnetization(self):
        return np.abs(np.sum(self.lattice)) / self.N

    def run(self, eq_steps=500, measure_steps=1000):
        T_c = 2.269
        if abs(self.T - T_c) < 0.3:
            eq_steps = max(eq_steps, 800)   # extra steps near T_c
        for _ in range(eq_steps):
            self.step()
        E_list, M_list = [], []
        for s in range(measure_steps):
            self.step()
            if s % 3 == 0:
                E_list.append(self.energy())
                M_list.append(self.magnetization())
        E_arr, M_arr = np.array(E_list), np.array(M_list)
        C_V = np.var(E_arr, ddof=1) / self.T**2
        chi = np.var(M_arr, ddof=1) / self.T
        return np.mean(E_arr), np.mean(M_arr), C_V, chi

    def run_with_averaging(self, eq_steps=500, measure_steps=1000, n_runs=3):
        results = [[], [], [], []]
        for run in range(n_runs):
```

```python
            np.random.seed(42 + run * 1000)
            self.lattice = np.random.choice([-1, 1], size=(self.L, self.L))
            for i, v in enumerate(self.run(eq_steps, measure_steps)):
                results[i].append(v)
        return [np.mean(r) for r in results]

# Main temperature scan
L = 20
temps = np.linspace(1.5, 4.0, 14)
data  = {'E': [], 'M': [], 'C': [], 'X': []}

for T in temps:
    model = IsingCellularAutomata(L=L, T=T, rule='glauber')
    E, M, C, X = model.run_with_averaging(
                    eq_steps=500, measure_steps=800, n_runs=3)
    data['E'].append(E); data['M'].append(M)
    data['C'].append(C); data['X'].append(X)
```

Listing 1: Core class and main simulation loop from `01_ising_cellular_automata.py`.

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